FactWorld: A Reproducible Instrument for Composing State Tracking and Recall

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Abstract

FactWorld is a benchmark suite that measures recall, state tracking, and their composition, identically for frontier models over an API and for small models trained from scratch. Every task must both differentiate frontier models and support architecture exploration in small models trained from scratch. Every number reproduces from committed scripts and can be checked with an API key or a single GPU. We find that the component abilities are largely solved at the frontier, but their composition is not. With reasoning off, most models hold little composition in their weights and architecture. With reasoning on, composition is bought by the token, and the price differs significantly by model. In task-specific training, each element of the composition requires a specific architectural or training capability.

1 Introduction

Recall and state tracking are usually benchmarked separately: recall with multi-query associative recall (Arora et al., 2023), state tracking with S_5 word problems (Barrington, 1989; Liu et al., 2023). Whether a model can compose them is a different question: track a changing state, then act through what it tracked. FactWorld measures that composition.

FactWorld is a frozen, versioned suite of tasks that render to natural language over a constrained vocabulary, with deterministic examples from fixed seeds. Gold answers come from a symbolic oracle applied to the underlying world state, never from parsing rendered text, so labels cannot leak. A validity gate certifies that no shallow shortcut succeeds on any task. One scoring rule and one protocol serve both frontier models over an API and small models trained from scratch, so a finding made in one regime can be checked in the other.

Measured this way, the component abilities are largely solved at the frontier, but their composition is not. With reasoning off, most models hold little composition in their weights and architecture, and the ordering does not follow the reasoning-on ranking. With reasoning on, composition is bought by the token, and the price differs significantly by model. In task-specific training, each element of the composition requires a specific architectural or training capability, and the composed task itself remains unsolved at the scales tested.

The instrument is deliberately narrow. Every task is single-turn and single-answer-span, with no tool use, planning, or multi-turn action, and the results are measurements within the regime tested, not scaling laws.

2 The FactWorld instrument

2.1 Tasks

Each task isolates one component ability or one composition of them (Table 1).

	task	role
Component: recall	`recall_copy_v1`	deferred-readout MQAR variant; pool breadth is the load axis
Component: recall (parametric)	`conflict_v1`	memorized map contradicted in context
Component: state tracking	`binding_v2`	last-write-wins over a give-stream
commutative variant	`commutative_v1`	per-entity accumulation mod k; order-free
non-abelian variant	`s5_v1`	order-sensitive role permutations
Composition: state × recall	`composite_copy_v2`	two-hop composition; §4.2 derives the gap statistic from it
Composition: recall ∘ recall	`chain_v2`	pointer chase at fixed breadth, explicit hop count
Composition: state × dereference	`s5_chain_v3`	retired: it does not separate the roster (§4.1)

Table 1: The task taxonomy: components first, then their compositions.

Composition (`composite_copy_v2`). A set of facts maps agents to values, and a stream of give events moves objects between agents. The query asks for the value of the agent that currently holds a given object:

```
g2's a0 is v70. g4's a0 is v24. g0's a0 is v109. g1's a0 is v48.
s0 gives o3 to g4. s1 gives o3 to g1. s2 gives o3 to g2.
what is a0 of the holder of o3?
gold: g2 v70 .
```

The model must resolve the holder by tracking the give-stream (last write wins), then look up that holder's value in the facts. The facts are resampled every example, so the value is read from context, not from weights. The answer is two tokens, a holder and a value, and each leg is scored independently: holder-right/value-wrong means the model tracked state but failed the recall hop, and the reverse means the opposite. This decomposition localizes failures throughout the report.

S_5 **(`s5_v1`).** A stream of swap and cycle events permutes the roles of a set of agents, and the query asks for one agent's final role. Swaps and cycles do not commute, so the running permutation must be carried step by step.

s5_chain (`s5_chain_v3`). This task composes the two stress components in one task. It is retired — §4.1 gives the measurement — and generable for reproduction. $k = 16$ agents hold an `a0` pointer map, initially one 16-cycle. L order-sensitive swap/cycle events permute the pointer values: the S_5-style state-tracking load. The query then dereferences the final map 8 hops deep (`what is a0 of a0 of ... of gX? (8 hops)`): the pointer-chase load. Neither component alone suffices. Every item is gated so that the query start sits on a final-map cycle longer than the query depth. The nine path nodes are therefore distinct, echoing the queried agent or any fixed hop scores exactly 0, difficulty is uniform across items, and chance is 1/16. Cycle events state all their assignments simultaneously, so no sequential misreading is available. Unit tests pin both the gate and the rendering.

Difficulty knobs (length L, pool breadth, chain depth) are calibration parameters, used to place each model class mid-scale.

2.2 Scoring

The metric is **match**: strip a trailing period from both sides and compare the model's first len(gold) whitespace tokens to the gold answer, binary per item, no partial credit. Containment is the one published diagnostic, and any cell (one model on one task at one setting) where containment exceeds match by 0.08 or more has its raw predictions read before the number is believed. When reasoning is on, match scores the answer a multi-line emission commits to (a single-token final line, an emphasized answer closing a statement, or a lone token in a trailing code fence), never the working's first tokens. The commitment is located structurally, never by demanding a rigid output format.

2.3 Regimes and floors

Every frontier model runs in two regimes. **Instant** disables reasoning under a hard one-line answer contract and a 96-token cap; it measures what the weights alone compute. **Thinking** grants a generous reasoning budget at the shared top effort setting; it measures what reasoning adds. The two rankings are near-orthogonal, so profiles are per-axis, never a single scalar.

Instant cells are read against two floors, recomputed at render time from the exact task items. The *recency heuristic* answers the last event's recipient and scores 0.04. The *object-filter floor* filters events to the queried object but guesses among its writes and scores $\mathbb{E}[1/w]$: 0.41 at L16, decaying roughly as $1/L$. A score near the floor shows object filtering, not state tracking.

The floors exist because samplers leak shortcuts. A give-stream sampler that draws events uniformly leaves the queried object's resolving write close to the end of the stream, so a one-line recency heuristic scores 0.33, indistinguishable from genuine mid-pack state tracking. The current sampler places the queried object's last write uniformly over the stream, which drives that heuristic to chance and exposes the object-filter floor as the real bar: the weakest models sit at or below 0.41 where they previously looked mid-pack. The validity gate checks the recency heuristic on every give-stream task.

3 Validating the instrument

Before the suite compares architectures or models, it must reproduce the field's established single-capability dissociations. Three of them reproduce on the natural-language format, three seeds each.

1-hop associative recall (MQAR). The value is read adjacent to the key. Attention solves it: gdp_hybrid 1.00, fprm 1.00, transformer 1.00 (pool 16).

Deferred readout recall. The value must be read out at an arbitrary later position; this is the regime that composition actually requires. Scores: gdp_hybrid 0.73, fprm 0.50, transformer 0.19 (pool 5). Every architecture aces 1-hop; only the recurrent hybrid solves deferred readout.

S_5 **length extrapolation under dense supervision.** Train dense, evaluate free-running past the training length (Table 2). The product recurrence extrapolates; the transformer and the looped block shortcut past the trained length.

arch	L16 (train)	L64 (4×)	L128 (8×)
gdp_hybrid	1.00	0.90	0.82
fprm	1.00	0.17	0.23
transformer	0.79	0.22	0.22

Table 2: S_5 length extrapolation under dense supervision, three seeds.

4 Benchmarking the frontier

We run thirteen frontier models through the instrument. Instant cells run $n = 100$ under the answer contract. Thinking cells run $n = 25$, with Wilson 95% intervals on every cell. At $n = 25$ one item is 0.04, and the ranking task resolves a broad band rather than a rank over most of the roster (§4.1). Models run at the shared top effort arm, `xhigh`, mapped down where the endpoint's ceiling is `high`. Truncated calls score wrong, and ^{t} marks a cell where at least one call hit its completion budget, so that cell reads as a lower bound. Cells that carry a contamination mark are excluded from orderings: ⊘ means not measurable at this budget (most calls truncate), † means covert or visible reasoning on an instant attempt, ‡ means the provider ignored the token cap, and ^{u} means unworked answers on a large fraction of calls, so the cell measures engagement, not capability. Three models (grok-4.5, muse-spark-1.1, claude-fable-5) are thinking-only: their endpoints cannot disable reasoning, so they carry no instant numbers.

4.1 FactWorldBench

`s5_chain_v3` was built as the ranking task: one number per model for how well it holds a mutating pointer map and then acts through it. It does not do that job and is retired. The L96 and L128 cells separate the tail from the rest of the roster, and the matched L64 cell prices completion tokens per call — nine of the thirteen models score 0.92 or better there, so their spend compares like for like — but no pair inside the top eleven separates at L96 (§7), so the cell orders the top of the roster by noise. `s5_chain_v4` blocks the backward walk v3 admits (the shallow adversary goes 1.000 to 0) and does not reach that defect: on the one model with clean cells at both versions it reads the same on each. What FactWorldBench publishes is the component and composition tasks; the composed cell that does separate is the composition instrument's, scored on the pad write rather than the answer (§4.6).

What the score resolves is the tail. At L96 no pair among the eleven models above qwen and sol separates (Fisher exact, all 55 pairs at $p \geq 0.05$), and Cochran's Q over those eleven returns $p = 0.777$: at $n = 25$ they are one undifferentiated band. Every one of the 18 separating pairs out of 78 involves qwen or sol, and sol's separation is engagement rather than capability (below). The roster mean is flat in length as well — 0.90, 0.89, 0.91, 0.90 at L32, L64, L96, L128 — so length moves individual models rather than the roster: L128 is where qwen halves to 0.44 while fable and gpt-5.5 hold 1.00.

Token spend does separate models the score does not. Over the nine models scoring 0.92 or better at L64, spend runs 5.0k per call (fable) to 18.0k (glm), a 3.6× range inside one score band. The ^{r} on nemotron's L96 marks a single rerun at a raised 98,304-token budget. At the planned budget it scored 0.84 with 12% truncation, a budget artifact rather than a capability limit; at L128 it holds 0.96 with no truncation.

The serving stack constrains the measurement. gpt-5.6-sol's Chat Completions shim caps the effort

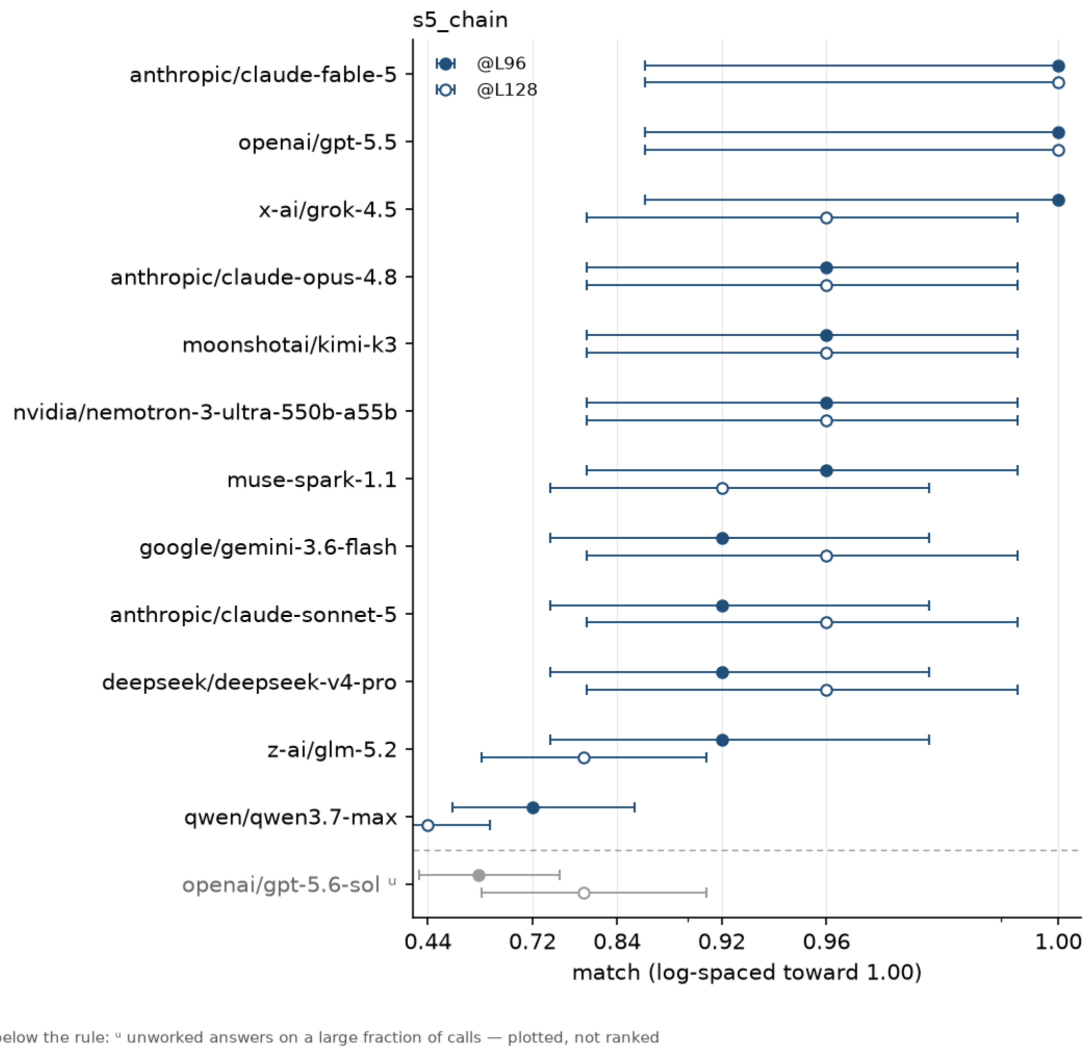

Figure 1: FactWorldBench scores at L96 (filled) and L128 (open) with Wilson 95% intervals, plotted on a log-spaced match axis. Overlapping intervals are not an ordering. Rows run in Table 3 order, by L96 then L128 then L64 spend; gpt-5.6-sol's cells carry ^{u}, take no part in that ordering, and are plotted greyed below the rule.

Model	s5_chain @L96	@L128	ctok/call @L64
anthropic/claude-fable-5	1.00	1.00	5014
openai/gpt-5.5	1.00	1.00	9343
x-ai/grok-4.5	1.00	0.96	7711
anthropic/claude-opus-4.8	0.96	0.96	9702
moonshotai/kimi-k3	0.96	0.96	10941
nvidia/nemotron-3-ultra	0.96^{r}	0.96	17071^{t}
muse-spark-1.1	0.96	0.92	12484
google/gemini-3.6-flash	0.92	0.96	8166
anthropic/claude-sonnet-5	0.92	0.96	12729
deepseek/deepseek-v4-pro	0.92	0.96	17052
z-ai/glm-5.2	0.92	0.80^{t}	17982^{t}
qwen/qwen3.7-max	0.72	0.44	12588
openai/gpt-5.6-sol	0.60^{u}	0.80^{u}	2444^{u}

Table 3: FactWorldBench: `s5_chain_v3` match at L96 and L128 ($n = 25$ per cell), and completion tokens per call on the matched L64 cell. ^{r} marks a single raised-budget rerun, ^{t} a cell with truncated calls (2 of 25 on each L64 cell, 1 of 25 at L128), and ^{u} a cell where the model returned unworked answers on a large fraction of calls.

ladder at `xhigh`, while its native Responses API exposes a further `max` level that lifts L96 from 0.60 to 0.88 at 2.3× the reasoning tokens; the scored rows stay on the shared `xhigh` setting for cross-model fairness.

Sol's `s5_chain` cells measure engagement rather than capability, which is why they carry ^{u}. Its per-call completion spend is bimodal with a literal gap: across the 300 archived `s5_chain_v3` calls, none spends between 294 and 1,136 completion tokens. In the low mode it emits about 150 tokens whatever the prompt size (per-length means 131 to 162), and all 109 low-mode calls answer with the 8-hop dereference of the *initial* pointer map — it answers the pointer chase and skips the event stream, an answer the rest of the roster gives on under 1% of its calls. That covers every one of sol's 101 wrong answers; the 8 low-mode calls that score are the items where the initial-map answer is also the gold one. Conditional on entering the high mode it is 191 for 191 across the same records. In the scored cells the low mode takes 68%, 40%, 44%, and 24% of calls at L32, L64, L96, and L128, so its score rises with prompt length because it disengages less often, not because it works harder.

The instrument's system prompt elicits the low mode. Every thinking cell in this report runs under one system prompt: "You are taking a short test. Answer each question with only the requested value or values, no explanation. Use the same spelling as in the question." Two of its clauses read as instructions to spend less effort. Three arms on the identical L64 items ($n = 25$, effort `xhigh`, 49,152-token budget) differ only in that prompt (Table 4): the canonical arm, no system prompt at all, and a neutral arm that keeps the answer-format contract and drops "a short test" and "no explanation".

Eight of the 25 items fell into the low mode under the canonical prompt; seven of those eight were worked under the neutral prompt, and no low-mode call in any arm is correct. The no-prompt arm's lower match is output format rather than composition: it works on 24 of 25 calls and all 24 reach the gold value, but three wrap it in math markup instead of committing it as text, which the

system prompt	match	containment	worked	match | worked	initial-map	ctok/call
canonical	0.68	0.68	0.68	1.00	0.33	3128
none	0.84	0.88	0.96	0.88	0.04	5942
neutral	0.96	0.96	0.96	1.00	0.04	4086

Table 4: gpt-5.6-sol on `s5_chain_v3` at L64, $n = 25$ deterministic items shared across the three arms, which differ only in the system prompt. *worked* is the fraction of calls above 500 completion tokens — low-mode calls run 86 to 258 tokens and worked calls start at 2,078, so the line's position inside that gap does not matter. *initial-map* is the fraction of answers that dereference the initial pointer map instead of the final one, counted under the eligibility rule every event-blind number in this report uses: an item counts only where the initial-map answer differs from the gold answer, which drops 1 of the 25 items from numerator and denominator alike.

committed-answer rule does not read; two of those three defeat the containment tokenizer as well, which is the 0.88 in the table. Keeping the answer contract and dropping the two clauses gives both, at 0.96 worked and 0.96 match on the same items.

The canonical arm's 0.68 reproduces the two archived L64 batches measured at the same effort (0.60 and 0.64, at 2,444 and 2,322 completion tokens per call); the third archived batch reads 0.84 at 4,966, twice the spend, and was measured at the vendor's higher level, so it is not a replicate of the other two. The low mode is therefore not a property of the model alone: dropping two clauses while holding the answer contract fixed moves the same items from 0.68 to 0.96 and the initial-map rate from 0.33 to 0.04. This is one model at one length with $n = 25$, and measures nothing about the twelve models the probe did not run. One consequence does reach the rest of the report: every scored thinking cell carries this system prompt, so the completion-token columns price spend under an instruction to answer without explanation.

4.2 The composed cell and the gap (instant regime)

With reasoning off, does the composition exist in weights at all? The composed cell is the two-hop query under the instant protocol at L16 and L64. The **gap**, binding@L16 − composed@L16, is interpretable only where the binding component is established. The recall half is free: every measurable model scores 0.97 to 1.00 across all recall variants, from the sanity check to the scaffolded probe, so the gap is a composition deficit, not a recall one (Table 5).

Where binding is solid the roster separates. Opus composes essentially for free: its gap of 0.06 is within instant test-retest variability. gpt-5.5 and kimi-k3 pay the largest clean deficits (gpt-5.5 binding 0.80, composed 0.46, gap +0.34; kimi-k3 gap +0.32), and at $n = 100$ the two are indistinguishable from each other. Both sit in the top band of the thinking ranking and hold among the least in-weights composition. For deepseek, qwen, and nemotron the binding leg's interval overlaps the 0.41 object-filter floor, so the gap renders $–^f$: floor minus floor is zero by construction, not a measurement.

4.3 Reasoning buys composition

On the composed cell at L16 ($n = 50$ per cell), reasoning effort gives a dose-response on the two models the sweep ran, kimi-k2.6 and glm-5.2 (Table 6; kimi-k2.6 is not on the benchmark roster). At effort none the holder leg reads against the 0.41 object-filter floor: object filtering, not established

Model	recall	binding @L16	composed @L16	composed @L64	gap @L16
anthropic/claude-opus-4.8	1.00	0.78	0.72	0.43	+0.06
anthropic/claude-sonnet-5	0.97	0.77	0.62†	0.32†	+0.15†
deepseek/deepseek-v4-pro	1.00	0.51	0.44	0.19	$-^f$
google/gemini-3.6-flash	1.00	0.69*	0.67*	0.26*	+0.02*
moonshotai/kimi-k3	1.00	0.65	0.33	0.29	+0.32
nvidia/nemotron-3-ultra	1.00	0.49	0.33	0.12	$-^f$
openai/gpt-5.5	1.00	0.80	0.46	0.33	+0.34
openai/gpt-5.6-sol	1.00	0.82	0.65	0.33	+0.17
qwen/qwen3.7-max	1.00	0.51	0.24	0.08	$-^f$
z-ai/glm-5.2	1.00	0.71	0.38†	0.13	+0.33†
recency heuristic (floor)		0.04	0.04	0.06	
object-filter floor		0.41	0.41	0.15	

Table 5: Instant-regime cells, match, $n = 100$. The gap is binding minus composed at L16. $-^f$ marks a gap that is not interpretable because binding sits at the object-filter floor; * marks a reasoning-off run served at effort minimal (the endpoint cannot disable reasoning).

composition. The first step off none carries the effect, and each further step is within noise of the one before it.

model		none	low	medium	high
kimi-k2.6	match	0.40	0.94	1.00	0.98
	containment	1.00	1.00	1.00	1.00
glm-5.2	match	0.22	0.74	0.74	0.80
	containment	0.40	0.96	1.00	1.00

Table 6: Reasoning-effort dose-response on the composed cell at L16, $n = 50$ per cell, with the containment diagnostic on the same cells.

Containment reads the two models apart. kimi-k2.6 has both answer tokens in its emission at every effort including none, so its entire dose-response is output format, not composition appearing. glm-5.2's containment climbs from 0.40 to 1.00, so its first step is composition appearing, and its residual match gap from low effort on is format. The lever is implicit reasoning: an explicit "write the holder, then the value" instruction hurts every model, including the reasoners that solve the composed cell under a plain prompt (format-fair ablation, $n = 100$).

4.4 The components under stress

The two state-stress components are read separately in the thinking regime, far past the composed cell's settings: chain d128 (a 128-hop pointer chase at fixed breadth $k = 257$, chance below 0.01) and S_5 @L256 (256 permutation events, concrete rendering, chance 0.20). The s5@128 ctok column is completion spend on the matched L128 cell that every model runs (Table 7).

The top half of the roster holds both components at or near ceiling, which is why the composite, not the components, carries the ranking. The budgets do not fully clear the reasoning these cells require. Nemotron truncates on 32% of its chain d128 calls at 16,384 tokens and on 20% of its S_5 calls at both L128 (16,384 tokens) and L256 (32,768), and kimi-k3 on 16% of its S_5 L256 calls;

Model	chain d128	s5 @L256	s5@128 ctok
anthropic/claude-fable-5	1.00	1.00	6405
openai/gpt-5.5	1.00	1.00	6989
google/gemini-3.6-flash	0.96	1.00	8234
muse-spark-1.1	1.00	1.00	9704
deepseek/deepseek-v4-pro	1.00	1.00	10043
anthropic/claude-sonnet-5	1.00	1.00	11866
anthropic/claude-opus-4.8	1.00	1.00	12683^t
openai/gpt-5.6-sol	0.88	0.92	2657
x-ai/grok-4.5	1.00	0.88	8069
qwen/qwen3.7-max	0.96	0.80	7904
moonshotai/kimi-k3	1.00	0.80^t	11355^t
nvidia/nemotron-3-ultra	0.60^t	0.80^t	12250^t
z-ai/glm-5.2	0.92^t	0.76^t	6282^t

Table 7: Thinking-regime component stress cells, match, $n = 25$: chain d128 (a 128-hop pointer chase, chance below 0.01), S_5 at L256 (256 permutation events, chance 0.20), and completion tokens per call on the matched L128 cell. ^{t} marks a cell with truncated calls, which score wrong: the cell is a lower bound on its score and its spend.

eight further cells here and in Table 3 truncate on one or two calls in 25. Every marked cell is a lower bound on its score and its spend. The chain query includes an explicit hop count because the bare nested phrase is a hop-counting confound at depth 128: models miscount the repetitions and stop a few hops short, a prompt-format artifact rather than a state failure.

The components also dissociate the regimes within one item set. The chain d16 cell runs in both regimes on identical items. In the thinking regime, the seven models measured on it all score 1.00. In the instant regime, every model that answers cleanly is at floor (best 0.08). A 16-hop pointer chase is serial work no roster model holds in weights, and every model that has been given room to work solves it.

4.5 The composition instrument

The composed cell whose floor is closed by a cost rule rather than by a list of shortcuts is `s5_bind_v3`: one event stream read through two structures — a pointer map rewritten by swaps and a holder map rewritten by last-write-wins gives — where every event resolves its second operand live through the map it does not write. Its two components are the same stream with the second operand named. Composing them is priced rather than asserted: the composed cell's cheapest correct algorithm costs 3.8–4.2× either component's at equal work, where every other task in the suite costs 1×.

The scored quantity is not the answer. Under a bounded per-event checkpoint the answer is generated from the model's own checkpoint, so a per-item perfect checkpoint makes that context byte-identical to the gold one and the answer registers the checkpoint and nothing else. What is scored is the single checkpoint token whose write is two dependent reads. Its floor is a maximum over the rows a class rule admits — a row may hold at most one structure, may chain at most one hop per emitted token, and may not cost more than the cell's own algorithm — and inside that rule the fixed-address family is swept exhaustively, over address spaces, block classes, both canonical indexings, token positions and typed map reads, with the winning member selected on a disjoint pool and scored held out.

Measured on `gpt-5.6-sol` at $k = 12$, $n = 24$ items: 1.000 at $L = 128$ and 0.758 ± 0.114 at $L = 256$ (12 of 24 items perfect) against floors of 0.247 and 0.255. Errors cluster within items — one derailed stream corrupts every later block — so the interval is taken over items, and a token-level interval understates it by about 4.4×. A cell costs \$3–5 per model, because on this read the model's reasoning is the scored output rather than an unscored overhead bought at an 8k–98k token budget and discarded.

4.6 Long context

Both regimes are stressed well past their calibration points. The thinking regime holds composition at long context: glm-5.2 holds 0.94 to 0.98 from L64 out to L1024 on the composed task ($k = 32$, pool 16), while its instant arm sits at or below the object-filter floor from L128 on. At this breadth, length is not the binding constraint for the thinking regime. S_5 under a concrete rendering with reasoning holds 0.90 at L128, degrading gradually (Appendix A).

5 Exploring the architectures

The same tasks supply training data for models trained from scratch with next-token prediction. Supervision is answer-only by default, with a staged curriculum for the composite flagship; dense per-step supervision appears only where it is the measured lever (the S_5 and commutative formation results). We train the following architectures:

- **transformer**: a standard decoder-only stack, the attention baseline.
- **gdn_pure / gdn_hybrid**: GatedDeltaNet (Yang et al., 2024), gated delta-rule linear attention; pure is attention-free, hybrid interleaves one full-attention layer per four.
- **gdp_pure / gdp_hybrid**: GatedDeltaProduct (Siems et al., 2025), the delta rule generalized to a product of Householder transformations per token ($n_h = 4$). This is the non-commutative recurrence the state-tracking results turn on; same pure/hybrid split.
- **fprm**: a weight-tied looped conv+attention block, a variant of the fixed-point looped transformer of Movahedi et al. (2026) (we did not run their model); one block applied repeatedly, so per-token FLOPs match the transformer at 5 to 11× fewer parameters.

Comparisons are matched on compute, not parameters, at budgets sufficient for the capable configuration to converge. At the flagship scale ($d_{\text{model}} = 768$, 8 layers), fprm's per-token FLOPs equal the transformer's (∼159 MFLOP) at 10M vs 76M params, and gdp_hybrid's Householder-product recurrence costs 1.28× the transformer's (204 MFLOP at 101M params).

5.1 The composite flagship

On the composed task, pool 16 at L16, with the staged curriculum (25k steps, 80k docs, 3 seeds, eval $n = 500$):

gdp_hybrid solves the binding leg and most of the recall leg, and is competitive with API models on this task despite training from scratch at 100M params. fprm solves binding perfectly and fails to recall the value of the holder it just resolved. The transformer fails both legs. The per-leg decomposition names the deficit: the failure is routing the resolved holder into the in-context recall lookup, the same deficit the frontier's instant gap measures.

arch	params	per-tok FLOPs	composite @L16	holder leg	value leg
gdp_hybrid	101M	204 MFLOP	0.833 ± 0.089	0.999	0.833
fprm	10M	159 MFLOP	0.109 ± 0.089	0.998	0.109
transformer	76M	159 MFLOP	0.001 ± 0.001	0.065	0.041

Table 8: The composed task trained from scratch under the staged curriculum, match at L16 with per-leg decomposition (3 seeds, eval $n = 500$).

5.2 Scale does not fix the composite

The same curriculum at three compute-matched sizes (2 seeds each):

arch	small (384×6)	medium (768×8)	large (1024×12)
gdp_hybrid	0.12 ± 0.08	0.73 ± 0.01	0.21 ± 0.21
fprm	0.12 ± 0.05	0.03 ± 0.01	0.03 ± 0.02
transformer	0.01 ± 0.00	0.01 ± 0.01	0.00 ± 0.00

Table 9: The composed task at three compute-matched sizes, match at L16 (2 seeds each).

Convergence is architecture-specific and scale-dependent: gdp_hybrid at 768×8 is the only cell whose seeds both converge (0.72 and 0.745). It does not carry to 1024×12, where the same architecture reads 0.42 and 0.00 — a cell whose mean is a seed lottery, not a score. The transformer floors at every scale including 202M params and 417 MFLOP/token, with containment near zero as well: a real floor, not a formatting miss. Wherever binding is solved and the composed cell fails, the value leg is what collapses; the routing deficit is scale-invariant.

5.3 Binding under breadth

Convergence on the binding leg is bimodal across seeds, so the rung reports p(converge), the fraction of seeds reaching 0.9 (3 seeds per cell, 45 runs at $d = 256$). fprm converges 3/3 at B6, 2/3 at B8, 1/3 at B12, 3/3 at B16, and 0/3 at B24. gdp_hybrid converges 0/3 at B6 and B8, 1/3 at B12, 1/3 at B16, and 2/3 at B24. The transformer converges 0/3 at every rung, per-seed 0.03 to 0.26. Within a cell the seeds sit at the extremes and not near their mean: gdp_hybrid at B24 reads 0.98, 0.04, 1.00, and fprm at B12 reads 1.00, 0.17, 0.75. Last-write state requires recurrence — the transformer fits at no breadth — but neither recurrent form shows a breadth ceiling at this training budget: p(converge) is not monotone in breadth for either, and a cell mean over three seeds of that shape is not a ranking.

5.4 Chain depth does not extrapolate

Trained at chain depths 2 and 3, no architecture clears the 1/6 guess at depths 4 and 5 by more than seed noise (fprm 0.21/0.15, transformer 0.16/0.09, gdp_hybrid 0.01/0.00; 3 seeds each, and fprm's depth-4 seeds are 0.245, 0.22, 0.165). gdp_hybrid fits the training distribution best and scores worst held-out: a depth-specific circuit, systematically wrong one hop out, not a guesser. Dense intermediate-hop supervision does not help (0.00 to 0.10 held-out). Over the API, frontier models solve the same composition at depth 16, but only in the thinking regime.

5.5 S_5 forms under dense supervision and survives weaning

Under answer-only supervision S_5 floors for every architecture. It forms when the training signal carries the state: interleave the oracle's state checkpoint every K events and evaluate free-running (10 seeds):

K (stride)	value @L16	value @L64	converge @L16
1 (dense)	1.00	0.75	10/10
2	0.98	0.40	10/10
4	0.19	0.20	0/10
8	0.21	0.20	0/10

Table 10: Supervision density for gdp_hybrid: the circuit forms at a checkpoint every ≤ 2 events and is gone below.

The threshold is sharp, and it is the threshold for gdp_hybrid. It is not architecture-independent. Every architecture forms the circuit under a checkpoint every event: at $K = 1$, gdp_hybrid scores 1.00 with 10/10 seeds converged, fprm 1.00 with 5/5, and the transformer 0.83 ± 0.25 with 8/10. At $K = 2$ the transformer is already at chance (0.22 ± 0.03, 0/10) where gdp_hybrid still converges 10/10; fprm has no $K = 2$ arm. A checkpoint every event forms the circuit in every architecture measured, and how far the checkpoints can be thinned is architecture-specific. Length extrapolation is a separate requirement (§3). The circuit also survives weaning to label-free deployment: train dense, fine-tune on mixed densities including answer-only, and 8/8 seeds converge free-running, extrapolating on par with dense-only. Stressed to 32× the trained length on the dense-supervised non-abelian composite, the recurrent hybrid holds about 0.5 out to L512 while the looped block stays at floor.

5.6 Commutative state

The commutative rung (per-entity accumulation, order-free) does not form under answer-only supervision at any measured setting: chance for every architecture at $d = 256$. Dense per-step traces form it in-distribution for the recurrent architectures (gdp_hybrid 0.82 ± 0.15, fprm 0.65 ± 0.26 at L16; transformer at chance), and no run carries it past the training lengths. The pattern matches S_5: supervision density enables formation, and extrapolation is a separate, unmet requirement.

6 What each element of the composition requires

Table 11 assigns each element of the composition the architectural or training capability that produced it in local training.

Depth extrapolation and the value leg of the composed cell remain unsolved at the scales tested: no measured architectural or training choice provides the first, and only one curriculum at one scale converges the second.

7 Discussion

The same composition probes that rank the frontier roster also separate architectures trained locally, and the per-leg decomposition links the regimes: a finding about routing the resolved holder into

element	requirement
adjacent (1-hop) recall	attention: every architecture aces adjacent readout
deferred recall	product recurrence: the transformer aces adjacent, fails deferred (0.19 vs 0.73)
last-write state	recurrence: both recurrent forms converge at some breadth and neither monotonically, p(converge) swinging 0/3 to 3/3 between adjacent rungs; the transformer converges at no breadth
non-abelian state (formation)	dense per-step supervision: every architecture forms it at a checkpoint every event; how far the checkpoints thin is architecture-specific (gdp_hybrid holds at every 2, the transformer is at chance there)
non-abelian state (extrapolation)	recurrent hybrid: gdp_hybrid 0.75 @L64; fprm and transformer collapse past train length
commutative state (formation)	dense per-step supervision plus recurrence; extrapolation does not follow
depth extrapolation	**open**: unsolved by every measured choice, with or without intermediate-hop traces
local composition (value leg)	**open** at the default recipe: only the staged curriculum at 768× 8 converges it

Table 11: What each element of the composition requires.

recall can be checked in both settings.

The main findings:

- **Composition is where frontier models still separate.** The components are largely solved in the thinking regime, but their composition is not: the ranking composite separates the tail by score and the rest of the roster by token spend. With reasoning off, the composed cell shows that most of the roster holds little composition in weights, with an ordering the thinking ranking does not predict: gpt-5.5 sits in the top band of one and pays among the largest gaps on the other.

- **Composition responds to reasoning.** Effort lifts the composed cell off the object-filter floor at the first step (0.22 to 0.74 for glm-5.2, 0.40 to 0.94 for kimi-k2.6) and the thinking regime holds it out to L1024. Explicit prompting does not substitute.

- **Non-abelian state tracking responds to reasoning only under a concrete rendering.** Neither reasoning over abstract tokens nor a concrete rendering without reasoning suffices; the combination solves it, with a model-dependent length limit (Appendix A).

- **For local models, S_5 forms under dense supervision.** A state checkpoint every event forms the circuit in every architecture measured; thinned to every two events it still forms in the recurrent hybrid, where it extrapolates in length and weans to label-free deployment, and is at chance in the transformer.

- **Architecture carries length generalization.** A learned state circuit generalizes in length only on a recurrent hybrid; transformers and looped blocks shortcut.

These are results within the regime tested, not scaling laws.

8 Limitations and related work

Limitations. The scale regime is bounded: $k = 5$ S_5, local models of 3M to 269M params matched on compute at 32 to 540 MFLOP/token, pretrained models from a few billion to about a trillion parameters. Composition is 2-hop throughout except the ranking task's 8-hop dereference. Instant cells run $n = 100$; thinking cells run $n = 25$ to 50 because API costs scale with reasoning tokens. At $n = 25$ the ranking task separates the tail and prices tokens, and it does not resolve the top of the roster: no pair inside the top eleven models separates at L96, so the instrument as run supports a two-band read plus a spend ordering, not a rank. Local architecture cells run 2 to 10 seeds and several are bimodal, so their means carry a seed lottery that p(converge) reports and a mean hides. The component-to-agent mapping is a motivating analogy, not a proven one.

Related work. Recall is grounded in multi-query associative recall (Arora et al., 2023). State tracking is grounded in the S_5 word-problem literature: Barrington (1989) makes S_5-recognition NC^1-complete; Merrill and Sabharwal (2023) and Merrill et al. (2024) bound transformers and SSMs within TC^0; Liu et al. (2023) show transformers learn depth-limited shortcuts; Grazzi et al. (2024) unlock state tracking in linear RNNs via negative eigenvalues; Siems et al. (2025) introduce the Householder-product recurrence our gdp models use; Yang et al. (2024) introduce GatedDeltaNet. FactWorld's contribution is measuring the components independently and composed, under one protocol, for API models and from-scratch models alike.

9 Reproducibility

Every claim maps to a committed script and raw results in the repository. The data, oracle, and eval layer is pure stdlib; training runs need one CUDA GPU. Raw per-cell records (all attempts, usage, diagnostics) are one JSON object per cell in an append-only history file; rendered benchmark pages carry per-cell Wilson intervals, contamination marks, figures, and provenance.

A S_5 at the frontier

Without reasoning, S_5 floors at every length. Models do real step-by-step tracking on individual examples, then stall: they report a role the queried agent held at a recent step rather than the final one. In aggregate this is chance at every length from L4 to L128, for every model measured.

With reasoning on, the outcome depends on the rendering. Reasoning over the abstract token rendering leaves a strong model near 0.33, and a concrete rendering (people and jobs) without reasoning scores at chance. Together they solve the task. Under an 8,192-token budget the concrete-plus-reasoning curve degrades gradually: glm-5.2 holds 1.00 at L32, 0.97 at L64, and 0.90 at L128, with residual errors that are genuine wrong answers, not truncation. The degradation point is model-dependent (kimi-k2.6 degrades sooner), and there is no abrupt break.

Frontier models solve S_5 by reasoning over a concrete rendering. Small models trained from scratch solve it when the training stream carries a state checkpoint at least every two events (§5.5).

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